

Import Data using Transform Maps

ServiceNow Project Documentation

Date	01 October 2026
Team ID	SWTID-2026-4498
Project Name	Import data using Transform Maps
Platform	ServiceNow (Personal Developer Instance - https://dev447766.service-now.com)
Maximum Marks	5 Marks

1. Introduction

Project Title: Import data using Transform Maps

Team Members:

S.No	Team Member	Role
1	Sarah Christa T	Team Leader & Project Coordination Lead
2	Saravanan K	Quality Assurance & Presentation Support
3	Sarudharshini B	Project Developer & Technical Documentation Lead
4	Sathya Priya R	Data Validation & Testing Lead
5	Selva Karthiga B	Reports, Dashboard & Documentation Support

2. Project Description

This project focuses on importing structured data from external spreadsheets (Excel / CSV) into the **ServiceNow** platform using **Import Sets** and **Transform Maps**. It simulates a real-world scenario in which bulk employee and employee-training data is received in spreadsheet format and has to be migrated into ServiceNow efficiently and accurately.

The spreadsheet data is first loaded into an Import Set table, which acts as a staging area. A Transform Map then maps the source fields to the appropriate target table fields. When the transformation is executed, records are created or updated in the target table and validated to ensure data integrity. The project uses **Coalesce on Employee ID** so that repeated or updated files change existing records instead of creating duplicates, and it builds **Reports** and an **Employee Analytics Dashboard** to give HR teams visibility into the imported data.

Two custom tables were created for the project: **Employee Test** (u_employee_test) for employee master data and **Employee Training Records** (u_employee_training_records) for training data that references the ServiceNow User table. Access to the data is controlled through roles such as **Hr Manager**.

Scenarios

Scenario 1: HR receives an Excel sheet (NMEmp_details.xlsx) containing Employee ID, Employee Name, Email, Department and Location. The sheet is loaded into the staging table u_employee_import using

Load Data. The first load processes 5 rows with 5 inserts and 0 errors, and the NMEmp_details Import transform map copies them into the Employee Test table.

Scenario 2: Training details (training name, completion date, status and employee) are imported into the Employee Training Records table using the Employee Transform map. The Employee column is a reference to the User table, so every training record is linked to a user record.

Scenario 3: An updated file (Employee_Update.xlsx) arrives with changed details for existing employees and a few new employees. In an earlier test, before coalesce was enabled, repeated loads created duplicate rows (Employee IDs S104082, S104085 and S104086 appeared twice). After Coalesce was enabled on Employee ID in the NMEmp_details Import map, existing employees were updated, new employees were inserted, and every Employee ID appears only once in the Employee Test table.

Scenario 4: HR managers need a quick summary of the imported data. Three reports on the Employee Test table (pie chart by department, bar chart by location and a list report) are shown together on the Employee Analytics Dashboard.

3. Project Overview

Purpose

- Automate the migration of spreadsheet data into ServiceNow tables using Import Sets and Transform Maps.
- Maintain data accuracy and integrity by mapping fields explicitly and avoiding duplicate records through Coalesce.
- Provide verification and insight through Transform History, Import Log, Reports and a Dashboard.

Key Features

S.No	Feature	Description
1	Custom Tables	Employee Test and Employee Training Records tables created with string, date, choice and reference columns.
2	Form Layout Configuration	Form of Employee Training Records arranged using split sections for a clean two-column view.
3	Spreadsheet Load (Import Set)	Excel files loaded through Load Data into the staging table u_employee_import.
4	Transform Map and Field Maps	Source fields mapped to target fields (NMEmp_details Import for Employee Test; Employee Transform for training data).
5	Transform Execution and History	Transform run on an import set; results reviewed in Transform History and Import Log.
6	Coalesce on Employee ID	Existing employees are updated and only new employees are inserted, so no duplicate records are created.
7	Reports	Pie chart (Employees by Department), bar chart (Employees by Location) and Employee List Report.
8	Dashboard and Roles	Employee Analytics Dashboard built with the in-line editor; Hr Manager role used for access.

4. Technical Architecture

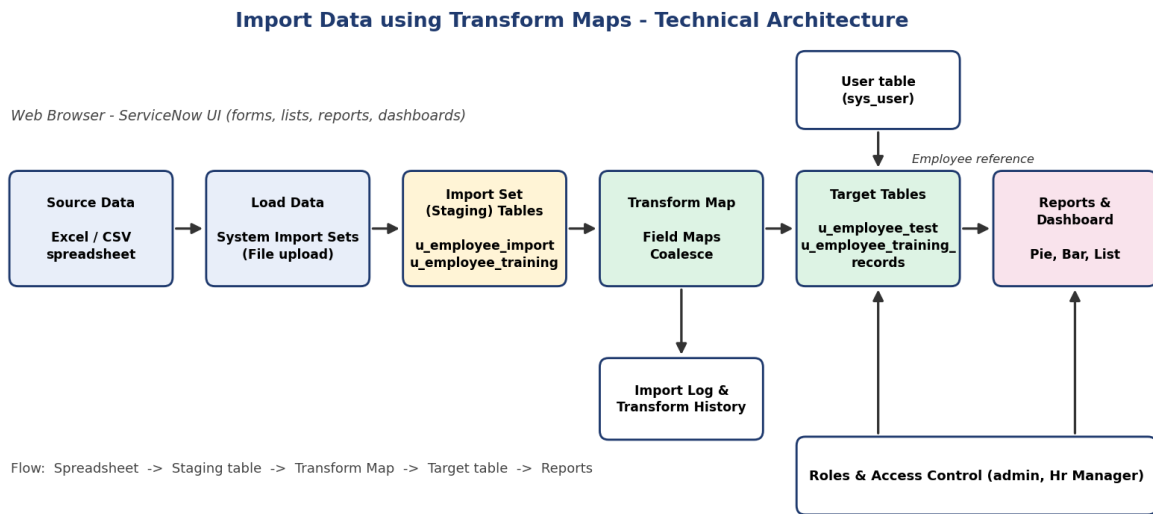


Figure 1: Technical architecture of the data import solution

Description: The solution follows a layered architecture built entirely on the ServiceNow platform.

Frontend: the ServiceNow web UI (forms, lists, Load Data page, Reports and Dashboards) is used by administrators and HR users in the browser. **Backend:** the Import Set and Transform engine reads the uploaded spreadsheet, writes rows to a staging table and applies the Transform Map (field maps and coalesce rules) on the server. **Database:** the platform database holds the staging tables, the target tables and the User table. Import Log and Transform History record the outcome of every run, and roles/ACLs protect the data.

ER Diagram

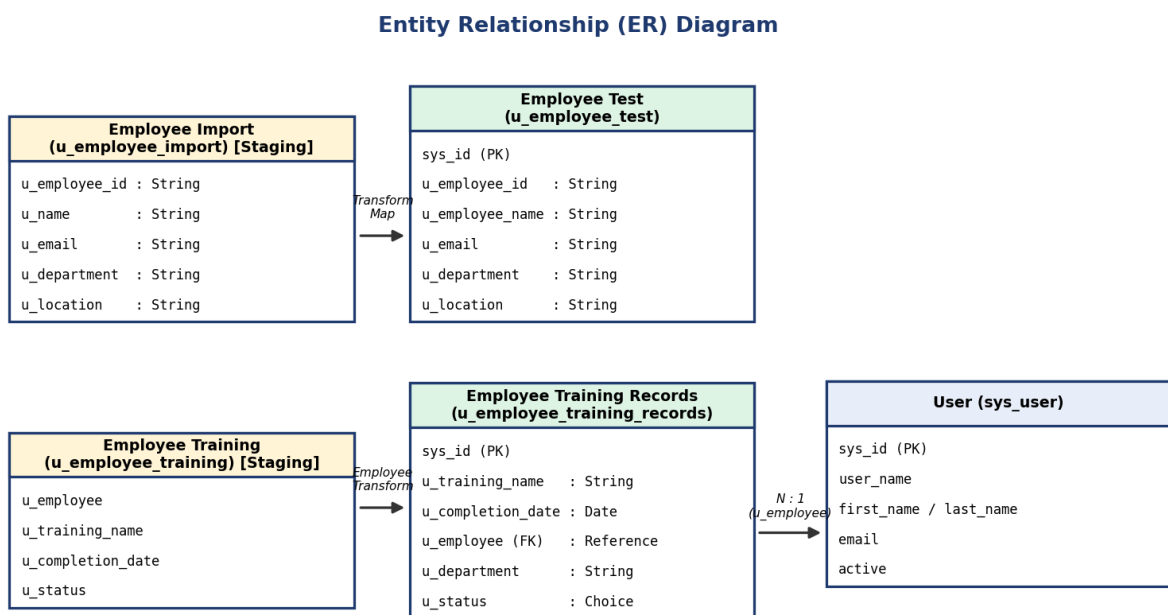


Figure 2: ER diagram of the staging, target and user tables

Description: Each staging table (Employee Import, Employee Training) is transformed into its target table (Employee Test, Employee Training Records) through a Transform Map. The Employee column of

Employee Training Records is a **reference field** to the User (sys_user) table, which gives a many-to-one relationship: many training records can belong to one user.

Table (Label)	Table Name	Purpose
Employee Import	u_employee_import	Staging table created by Load Data from the Excel files.
Employee Test	u_employee_test	Target table for employee master data; used for reports.
Employee Training	u_employee_training	Staging table for training data (source of Employee Transform).
Employee Training Records	u_employee_training_records	Target table for training records with a reference to sys_user.
User	sys_user	Out-of-the-box table of ServiceNow users.

5. Pre-requisites

- **ServiceNow Developer Account and Personal Developer Instance (PDI):** [ServiceNow Developer Program](#)
- **Knowledge of Import Sets and Transform Maps:** [ServiceNow Documentation](#)
- **Spreadsheet tool:** Microsoft Excel or Google Sheets to prepare the .xlsx source files (NMEmp_details.xlsx and Employee_Update.xlsx).
- **Administrator access:** the admin role to create tables, import data, build reports and create dashboards.
- **Basics of Reports and Dashboards:** Platform Analytics reporting concepts.
- **Web browser:** a current version of Google Chrome.

6. Setup Instructions

Prerequisites

No software installation, package dependencies or environment variables are required. Everything runs inside the ServiceNow instance through the browser.

Installation

1. Sign up at the ServiceNow Developer site and request a Personal Developer Instance.
2. Log in to the instance (<https://dev447766.service-now.com>) as an administrator.
3. Prepare the sample spreadsheets and keep the .xlsx files ready on your computer.
4. Create the custom tables (Milestone 2) and, if required, the user and role (Milestone 1).
5. Continue with Load Data, Transform Map, Coalesce, Reports and Dashboard as described in the milestones.

7. Folder (Artifact) Structure

ServiceNow does not use project folders, so the structure below lists the platform artifacts created for this project.

```
Import data using Transform Maps (Global scope)
|-- Source files
|   |-- NMEmp_details.xlsx ..... original employee data
|   '-- Employee_Update.xlsx ..... changed and new employees
|-- Tables
|   |-- Employee Test ..... u_employee_test
|   '-- Employee Training Records .... u_employee_training_records
|-- Import Set (Staging) Tables
|   |-- Employee Import ..... u_employee_import
|   '-- Employee Training ..... u_employee_training
|-- Transform Maps
|   |-- NMEmp details Import (Employee Import -> Employee Test)
|   |   Coalesce = true on Employee ID
|   |-- Employee_Update Import (Employee Import -> Employee Test)
|   '-- Employee Transform (Employee Training -> Employee Training Records)
|-- Roles and Users
|   |-- Role: Hr Manager
|   '-- User: backup user
|-- Reports (on Employee Test)
|   |-- Employees by Department (Pie)
|   |-- Employees by Location (Bar)
|   '-- Employee List Report (List)
'-- Dashboard: Employee Analytics Dashboard (in-line editor)
```

8. Project Workflow

Milestone	Activities
Milestone 1: Environment Setup and Data Preparation	Activity 1.1: Access the ServiceNow developer instance Activity 1.2: Prepare the sample spreadsheets Activity 1.3: Review the Hr Manager role and user
Milestone 2: Create Custom Tables	Activity 2.1: Create the Employee Test table Activity 2.2: Create the Employee Training Records table Activity 2.3: Configure the form layout Activity 2.4: Verify the form
Milestone 3: Load Data into Import Set	Activity 3.1: Load the spreadsheet using Load Data Activity 3.2: Verify the load result
Milestone 4: Create and Run the Transform Map	Activity 4.1: Create the Transform Map Activity 4.2: Review the field maps Activity 4.3: Run the transform Activity 4.4: Verify the Transform History
Milestone 5: Enable Coalesce and Update Data	Activity 5.1: Identify duplicate records Activity 5.2: Enable Coalesce on Employee ID Activity 5.3: Load the update file Activity 5.4: Run the coalesce-enabled transform map Activity 5.5: Verify updates and inserts
Milestone 6: Create Reports	Activity 6.1: Employees by Department (pie) Activity 6.2: Employees by Location (bar) Activity 6.3: Employee List Report
Milestone 7: Create the Dashboard	Activity 7.1: Create the dashboard with the in-line editor Activity 7.2: Add the reports to the dashboard Activity 7.3: Arrange and save the dashboard

9. Project Implementation

Milestone 1: Environment Setup and Data Preparation

This milestone prepares the instance, the source data and the access roles needed for the import.

Activity 1.1: Access the ServiceNow Developer Instance

1. Sign in to the ServiceNow Developer site and open the Personal Developer Instance.
2. Wake the instance if it is hibernating, then log in as the administrator.
3. Confirm that the ServiceNow home page and the All menu (application navigator) are available.

Activity 1.2: Prepare the Sample Spreadsheets

1. Create a new spreadsheet in Excel or Google Sheets.
2. Add the header row in row 1 and the employee records below it.
3. Save or download the file in **.xlsx** format. Two files are used: **NMEmp_details.xlsx** (original data) and **Employee_Update.xlsx** (changed and new employees).

The employee sheet uses the following columns. A few records from the original sheet are shown below:

Employee ID	Employee Name	Email	Department	Location
S104082	developer1	developer1@gmail.com	ServiceNow	TamilNadu
S104085	developer4	developer4@gmail.com	ServiceNow	TamilNadu
S104086	developer5	developer5@gmail.com	ServiceNow	TamilNadu

Activity 1.3: Review the Hr Manager Role and User

The **Hr Manager** role (Application: Global) is used to control who can work with employee data. The role record is shown below.

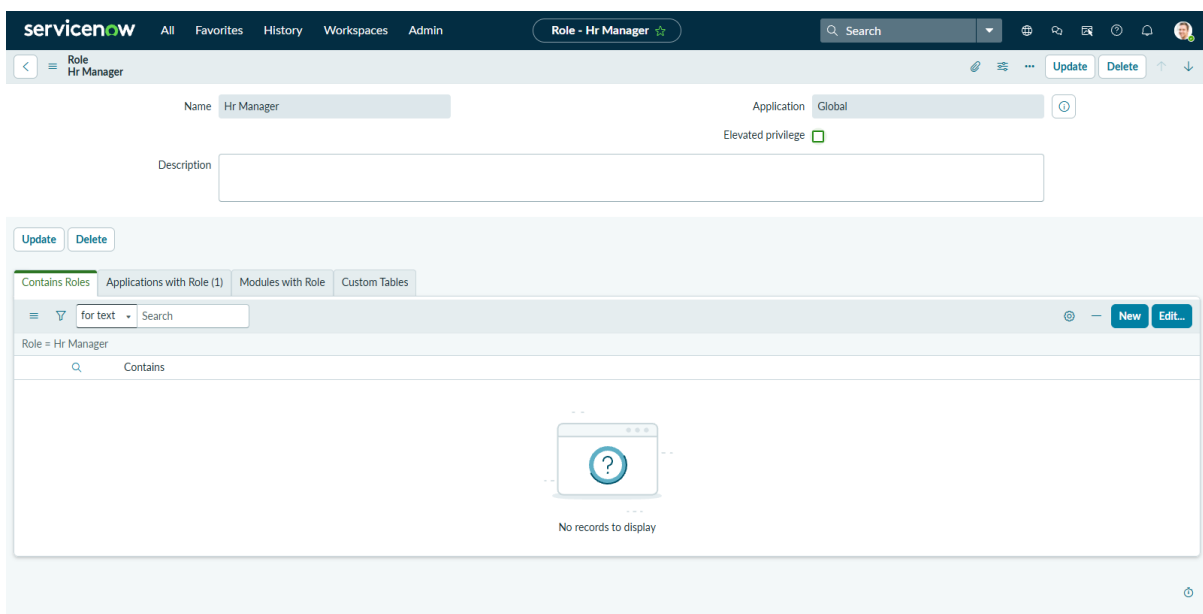


Figure 3: Role record - Hr Manager

A user record (User ID: **backup user**, Active, Identity type: Human) holds the Hr Manager role. The Roles tab lists Hr Manager (assigned directly) along with snc_required_script_writer_permission (inherited).

The screenshot shows the ServiceNow 'User' record for 'backup user'. The user is active, has an identity type of 'Human', and is assigned the 'Hr Manager' role. The 'Roles' tab is selected, showing a table of assigned roles.

Role	State	Inherited	Inheritance C
snc_required_script_writer_permission	Active	true	
Hr Manager	Active	false	

Figure 4: User record with assigned roles

Milestone 2: Create Custom Tables

Two custom tables store the imported data: Employee Test (employee master data) and Employee Training Records (training data).

Activity 2.1: Create the Employee Test Table

1. Open **All > System Definition > Tables** and click **New**.
2. Enter **Label:** Employee Test and **Name:** u_employee_test.
3. Add the columns Employee ID, Employee Name, Email, Department and Location, each of type **String**.
4. Click **Submit** to create the table.

Activity 2.2: Create the Employee Training Records Table

1. Open **System Definition > Tables** and click **New**.
2. Enter **Label:** Employee Training Records and **Name:** u_employee_training_records.
3. Add the columns shown in Figure 5: Training Name (String), Completion Date (Date), Employee (Reference to the User table), Department (String) and Status (Choice).
4. Click **Submit**.

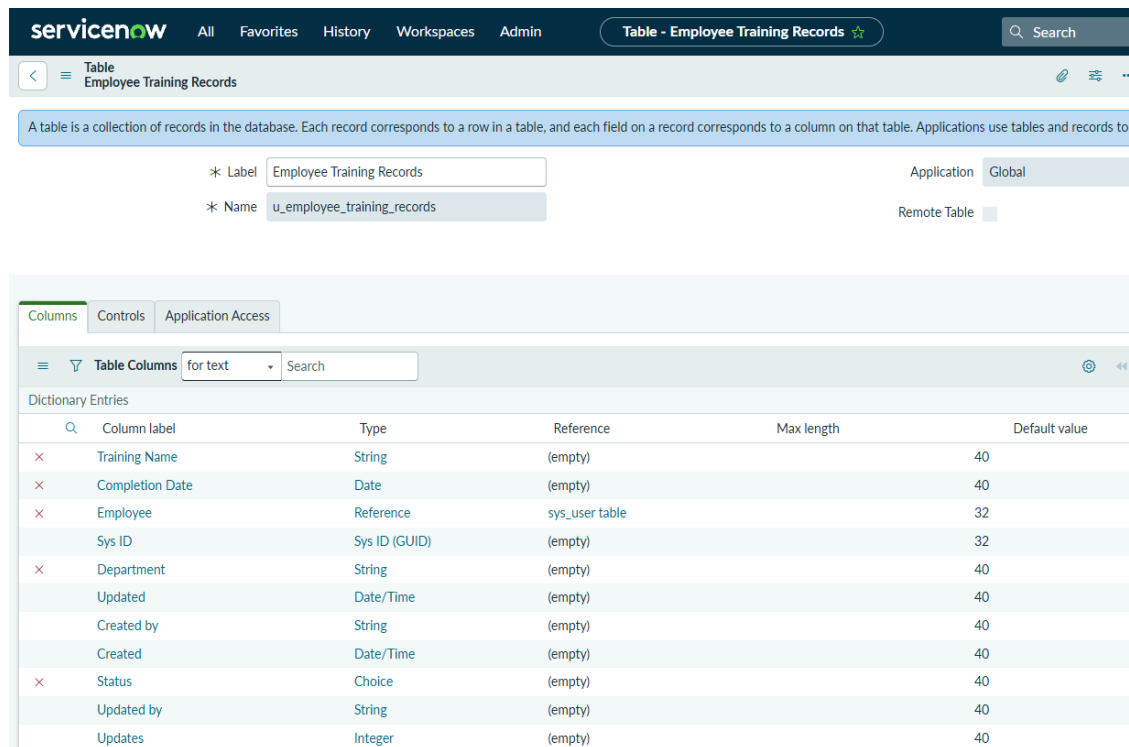


Table - Employee Training Records

Label: Employee Training Records
Name: u_employee_training_records

Application: Global
Remote Table: ☐

Columns Controls Application Access

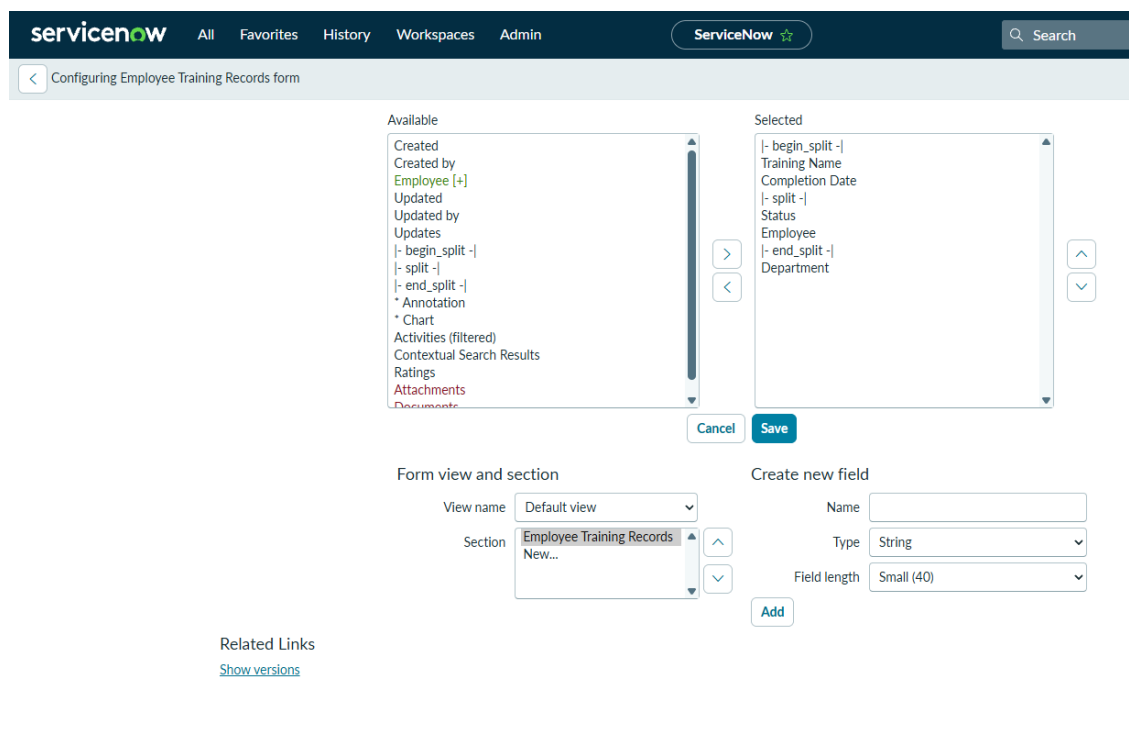
Table Columns for text Search

Column label	Type	Reference	Max length	Default value
Training Name	String	(empty)	40	40
Completion Date	Date	(empty)	40	40
Employee	Reference	sys_user table	32	32
Sys ID	Sys ID (GUID)	(empty)	32	32
Department	String	(empty)	40	40
Updated	Date/Time	(empty)	40	40
Created by	String	(empty)	40	40
Created	Date/Time	(empty)	40	40
Status	Choice	(empty)	40	40
Updated by	String	(empty)	40	40
Updates	Integer	(empty)	40	40

Figure 5: Employee Training Records table with its dictionary entries (columns)

Activity 2.3: Configure the Form Layout

1. Open a record of the table, click the **Form Context Menu** (three lines) and select **Configure > Form Layout**.
2. Choose **Form view**: Default view and **Section**: Employee Training Records.
3. Move the fields from **Available** to **Selected** in this order: begin_split, Training Name, Completion Date, split, Status, Employee, end_split, Department.
4. Click **Save**.



Configuring Employee Training Records form

Available

- Created
- Created by
- Employee [+]
- Updated
- Updated by
- Updates
- begin_split -
- split -
- end_split -
- * Annotation
- * Chart
- Activities (filtered)
- Contextual Search Results
- Ratings
- Attachments
- Documents

Selected

- begin_split -
- Training Name
- Completion Date
- split -
- Status
- Employee
- end_split -
- Department

Form view and section

View name: Default view

Section: Employee Training Records

Create new field

Name:

Type: String

Field length: Small (40)

Related Links

[Show versions](#)

Figure 6: Configuring the Employee Training Records form layout

Activity 2.4: Verify the Form

Open the table form and verify that Training Name and Completion Date appear on the left, Status and Employee on the right, and Department below. The sample record has Training Name "Full Stack Java", Status "Inprogress", Completion Date 2024-12-31 and Department "Product Management".

The screenshot shows the ServiceNow interface for 'Employee Training Records'. The header includes the ServiceNow logo and navigation tabs: All, Favorites, History, Admin. The main title is 'Employee Training Records - Created 2026-09-28 03:05:45'. Below the title, there are search and action buttons (Update, Delete). The form fields are arranged as follows: Training Name (Full Stack Java), Completion Date (2024-12-31), Status (Inprogress), Employee (empty), and Department (Product Management). At the bottom left, there are 'Update' and 'Delete' buttons.

Figure 7: Employee Training Records form after the layout configuration

Milestone 3: Load Data into the Import Set Table

Activity 3.1: Load the Spreadsheet using Load Data

1. Open **All > System Import Sets > Load Data**.
2. Select **Import set table: Create table**, enter **Label:** Employee Import. The **Name** u_employee_import is populated automatically.
3. Select **Source of the import: File** and choose NMEmp_details.xlsx.
4. Enter **Sheet number:** 1 and **Header row:** 1.
5. Click **Submit**.

To load further data into the same staging table, select **Existing table** and choose Employee Import [u_employee_import].

Activity 3.2: Verify the Load Result

After submission the progress page shows the ImportProcessor with State **Complete** and Completion code **Success**. In this run the message reads: Processed 5, inserts 5, updates 0, errors 0, empty and ignored 0. The Next steps links (Import sets, Loaded data, Create transform map, Import log) allow the next actions.

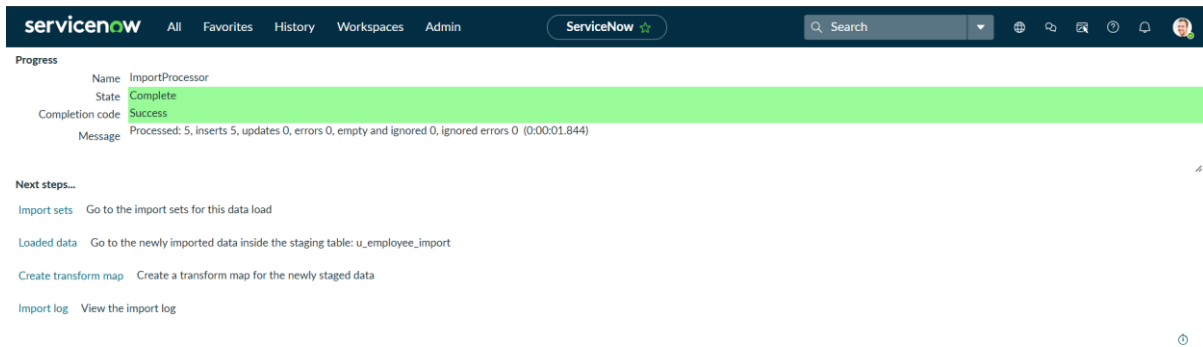


Figure 8: Import completed successfully - 5 rows processed, 5 inserted

Milestone 4: Create and Run the Transform Map

Activity 4.1: Create the Transform Map

1. Click **Create transform map** on the progress page, or open **System Import Sets > Administration > Transform Maps** and click **New**.
2. Enter **Name:** Employee Transform.
3. Select **Source table:** Employee Training [u_employee_training] and **Target table:** Employee Training Records [u_employee_training_records].
4. Keep **Active** and **Run business rules** selected, **Enforce mandatory fields:** No, **Order:** 100, then save the map.

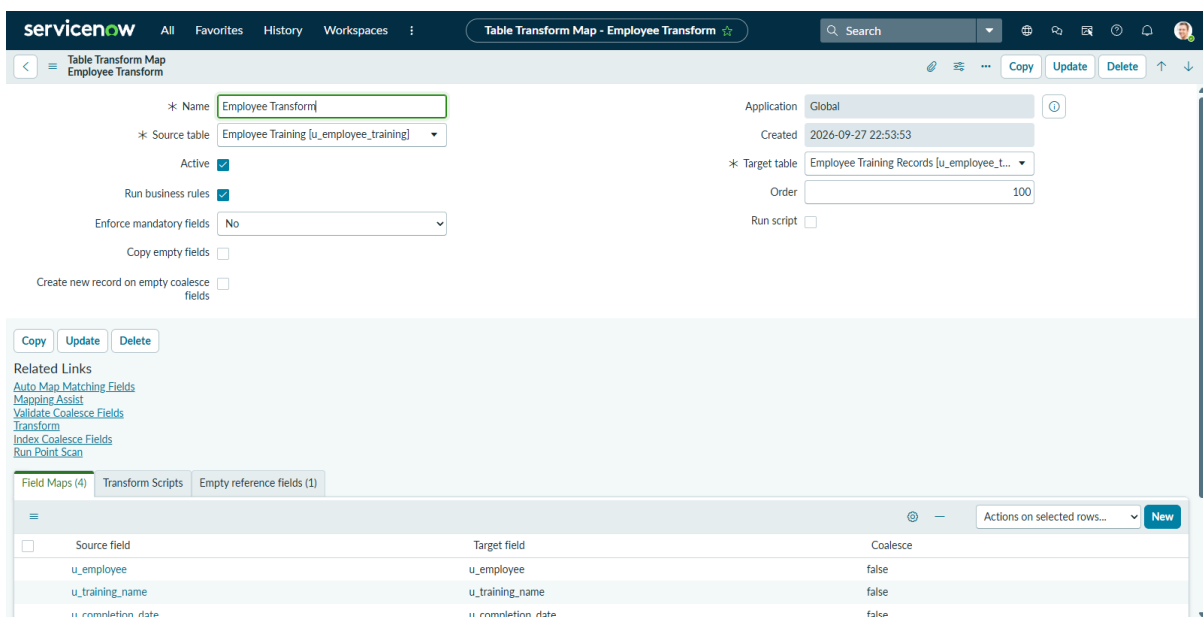


Figure 9: Transform Map - Employee Transform (source and target tables)

The same procedure creates the map for the employee data: **NMEmp_details Import**, with source table Employee Import [u_employee_import] and target table Employee Test [u_employee_test].

Activity 4.2: Review the Field Maps

Use the related link **Auto Map Matching Fields** (or **Mapping Assist**) to map source columns to target columns that have the same name. The Field Maps tab of Employee Transform shows four mappings:

Source field	Target field	Coalesce
u_employee	u_employee	false
u_training_name	u_training_name	false
u_completion_date	u_completion_date	false
u_status	u_status	false

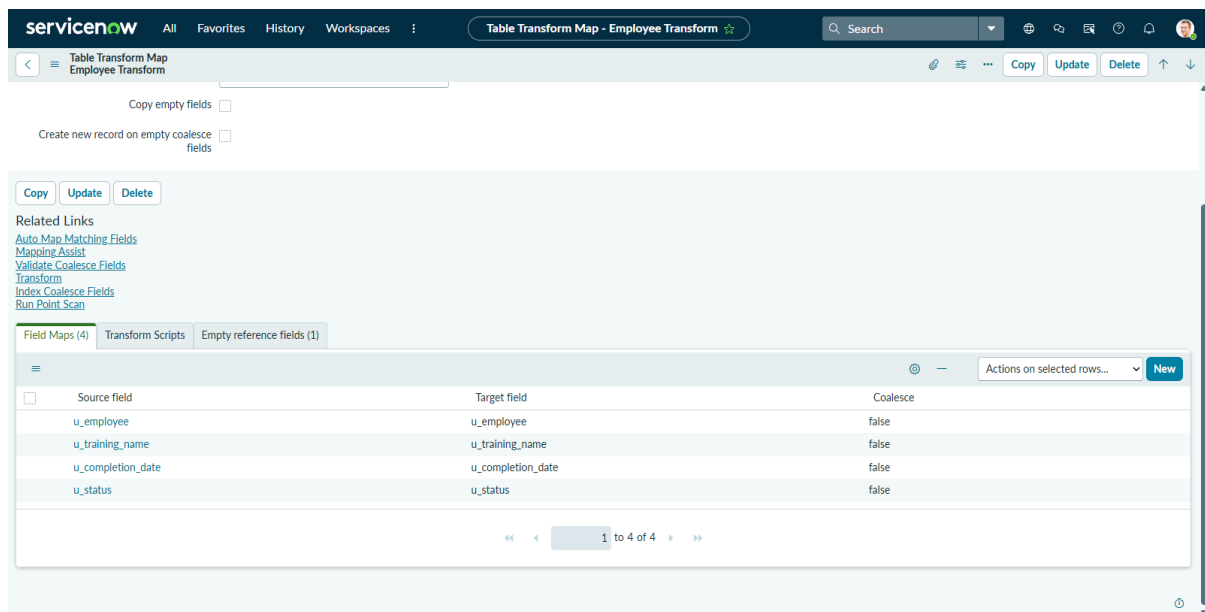


Figure 10: Field Maps (4) of the Employee Transform map

Activity 4.3: Run the Transform

1. On the Transform Map form, click the related link **Transform**.
2. Select the **Import set** (for example ISET0010002) and make sure the map appears under **Selected maps, run in order**.
3. Click **Transform** and wait for the progress page.

The progress page for "Transforming: ISET0010002" shows State **Complete**, Completion code **Success** and the message "Transformation complete". The links Transform history and Import log show the details of the run.

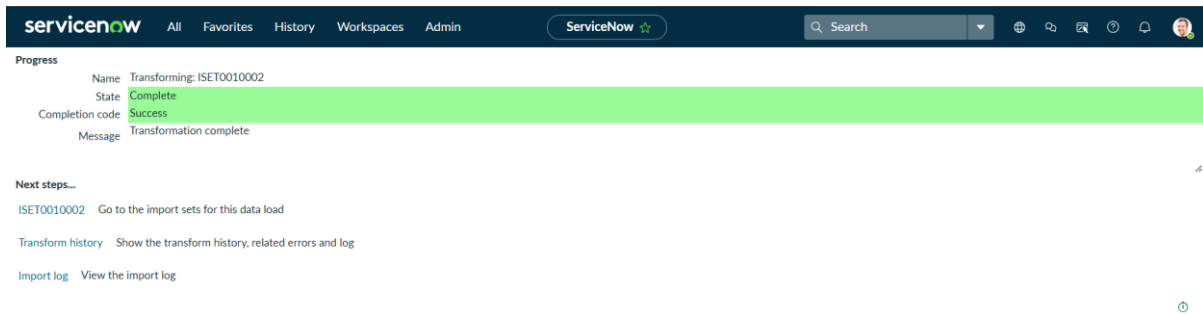


Figure 11: Transformation of import set ISET0010002 completed successfully

Activity 4.4: Verify the Transform History

Click **Transform history** to see the result of each run. For import set ISET0010009 (Employee Import [u_employee_import], transform map NMEmp_details Import) the history shows State Complete, Total 5, Inserts 5, Updates 0, Ignored 0, Skipped 0 and Errors 0 (Figure 12).

Started	State	Completed	Run time	Set	Import set table	Total	Inserts	Updates	Ignored	Skipped	Errors	Transform Map
2026-10-02 01:37:03	Complete	2026-10-02 01:37:03	0 Seconds	ISET0010009	Employee Import [u_employee_import]	5	5	0	0	0	0	NMEmp_details Import

Figure 12: Transform History - import set ISET0010009, map NMEmp_details Import (Total 5, Inserts 5, Errors 0)

Milestone 5: Enable Coalesce and Update Data

In real projects the same employee data is received more than once, and some of it has changed. **Coalesce** tells the transform which field identifies a record. If a record with the same value already exists it is **updated**; if not, a new record is **inserted**.

Activity 5.1: Identify Duplicate Records

In an earlier test, the same data was loaded more than once while Coalesce was false. The Employee List Report (Figure 13) then showed 11 records, with Employee IDs S104082, S104085 and S104086 each appearing twice. This is the problem that Coalesce solves.

Employee ID	Employee Name	Department	Email	Location
S104085	developer4	ServiceNow	developer4@gmail.com	TamilNadu
S104086	developer5	ServiceNow	developer5@gmail.com	TamilNadu
S104086	developer5	ServiceNow	developer5@gmail.com	TamilNadu
S104082	developer1	ServiceNow	developer1@gmail.com	TamilNadu
S104082	developer1	ServiceNow	developer1@gmail.com	TamilNadu
S104085	developer4	ServiceNow	developer4@gmail.com	TamilNadu

Figure 13: Employee List Report before coalesce - duplicate Employee IDs (11 records)

Activity 5.2: Enable Coalesce on Employee ID

1. Open **System Import Sets > Administration > Transform Maps** and open **NMEmp_details Import**.
2. In the **Field Maps** related list, open the map for Employee ID (u_employee_id).
3. Set **Coalesce** to **true** and click **Update**.

Source field	Target field	Coalesce
u_email	u_email	false
u_name	u_employee_name	false
u_department	u_department	false
u_location	u_location	false
u_employee_id	u_employee_id	true

Figure 14: Coalesce set to true on the Employee ID field map of NMEmp_details Import

The Field Maps (5) tab of NMEmp_details Import maps u_email to u_email, u_name to u_employee_name, u_department to u_department, u_location to u_location and u_employee_id to u_employee_id. Coalesce is true only for u_employee_id (Figure 14).

Activity 5.3: Load the Update File

1. Open **All > System Import Sets > Load Data**.
2. Select **Import set table: Existing table** and choose Employee Import [u_employee_import].
3. Select **Source of the import: File** and choose Employee_Update.xlsx, which contains changed details for existing employees and a few new employees.
4. Enter **Sheet number: 1** and **Header row: 1**, then click **Submit**.

Activity 5.4: Run the Coalesce-Enabled Transform Map

1. On the progress page, click **Run Transform** (or open the transform map and click **Transform**).
2. Select the new **Import set** (ISET0010010).
3. Make sure only **NMEmp_details Import - u_employee_test** is in the **Selected maps, run in order** box. Leave **Employee_Update Import** in the Available maps box (Figure 15).
4. Click **Transform**.

Note: two maps exist for the same staging table. If both were selected, every row would be processed twice, so only the map with Coalesce enabled is run.

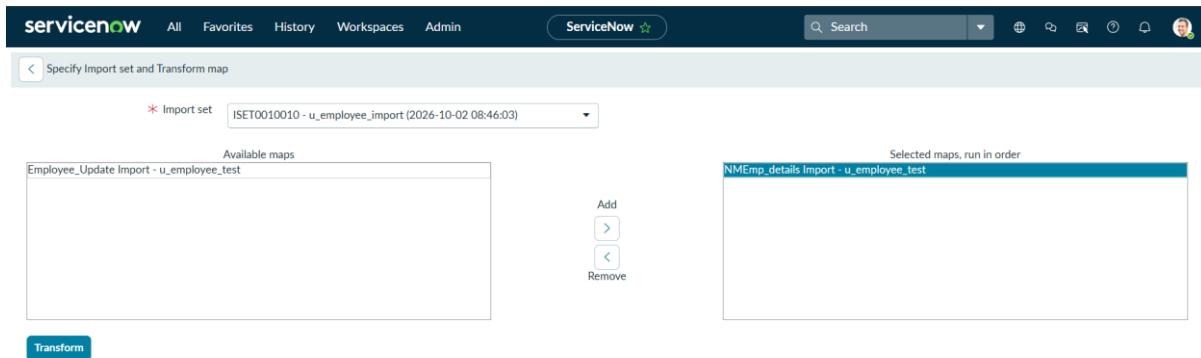


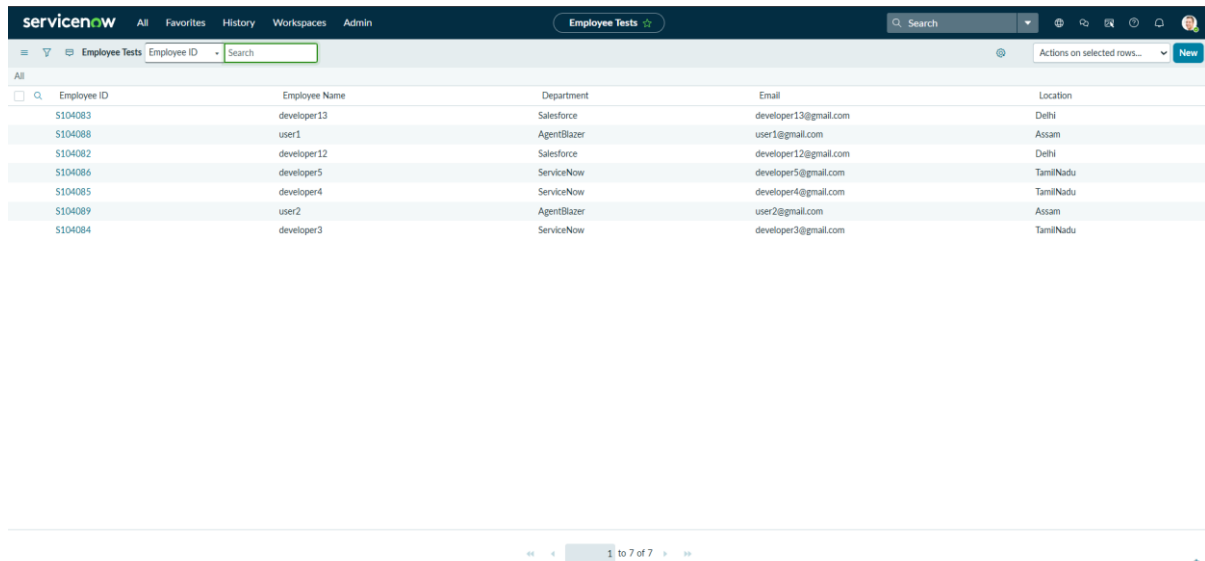
Figure 15: Run Transform - import set ISET0010010 with only NMEmp_details Import selected

Activity 5.5: Verify Updates and Inserts

1. Open **Transform history** and check the **Inserts** (new employees) and **Updates** (existing employees) columns.
2. Open **Employee Test > Employee Tests** and verify the data.

The Employee Tests list (Figure 16) now shows 7 of 7 records, and each Employee ID appears only once. For example, S104082 earlier showed developer1 (ServiceNow, TamilNadu) and now shows developer12 (Salesforce, Delhi), and the new employees S104088 and S104089 were added. No duplicate records were created.

Employee ID	Employee Name	Email	Department	Location
S104082	developer12	developer12@gmail.com	Salesforce	Delhi
S104083	developer13	developer13@gmail.com	Salesforce	Delhi
S104084	developer3	developer3@gmail.com	ServiceNow	TamilNadu
S104085	developer4	developer4@gmail.com	ServiceNow	TamilNadu
S104086	developer5	developer5@gmail.com	ServiceNow	TamilNadu
S104088	user1	user1@gmail.com	AgentBlazer	Assam
S104089	user2	user2@gmail.com	AgentBlazer	Assam



Employee ID	Employee Name	Department	Email	Location
S104083	developer13	Salesforce	developer13@gmail.com	Dehi
S104088	user1	AgentBlazer	user1@gmail.com	Assam
S104082	developer12	Salesforce	developer12@gmail.com	Dehi
S104086	developer5	ServiceNow	developer5@gmail.com	TamilNadu
S104085	developer4	ServiceNow	developer4@gmail.com	TamilNadu
S104089	user2	AgentBlazer	user2@gmail.com	Assam
S104084	developer3	ServiceNow	developer3@gmail.com	TamilNadu

Figure 16: Employee Tests after the update - 7 of 7 records, each Employee ID once

Milestone 6: Create Reports

Three reports are created on the Employee Test table.

1. Open **All**, search **Reports** and click **Reports** under **Core UI artifacts**.
2. Click **New** to open the report builder.

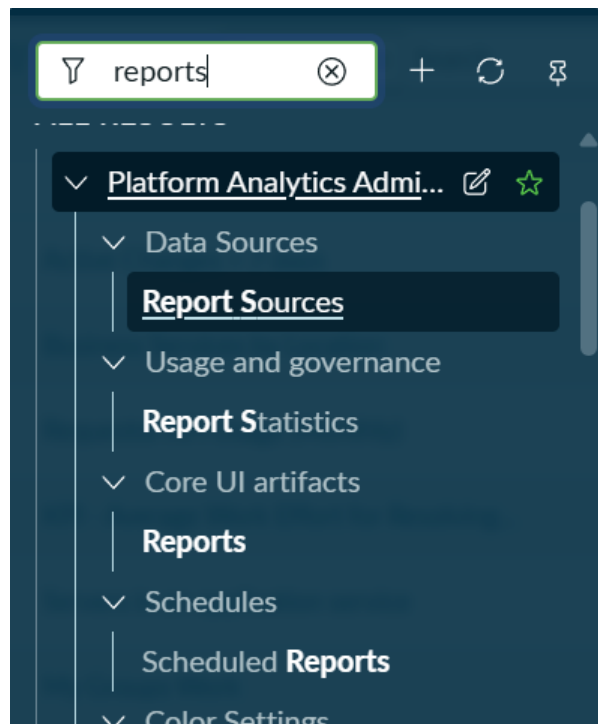


Figure 17: Reports module in the navigator (Core UI artifacts)

Activity 6.1: Report 1 - Employees by Department (Pie Chart)

1. On the **Data** tab enter **Report name:** Employees by Department, **Source type:** Table and **Table:** Employee Test [u_employee_test], then click **Next**.
2. On the **Type** tab select **Pie** and click **Next**.
3. On the **Configure** tab select **Group by:** Department and **Aggregation:** Count, then click **Next**.

- On the **Style** tab keep Chart color: Use color palette (Default UI14), tick Display data labels, then click **Run** and **Save**.

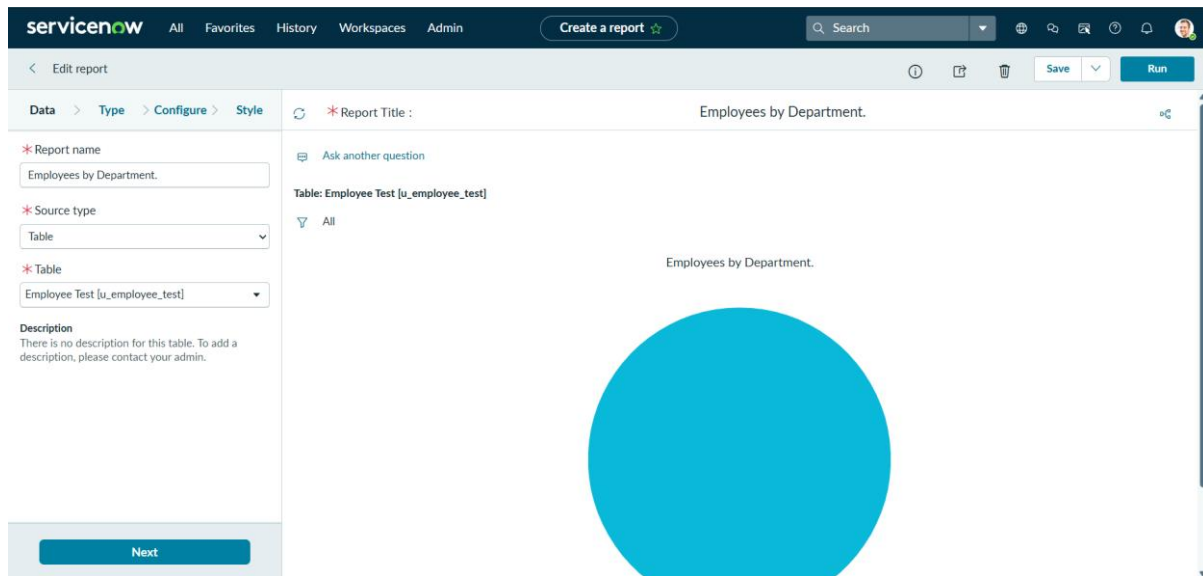


Figure 18: Report creation - Data tab (Employees by Department)

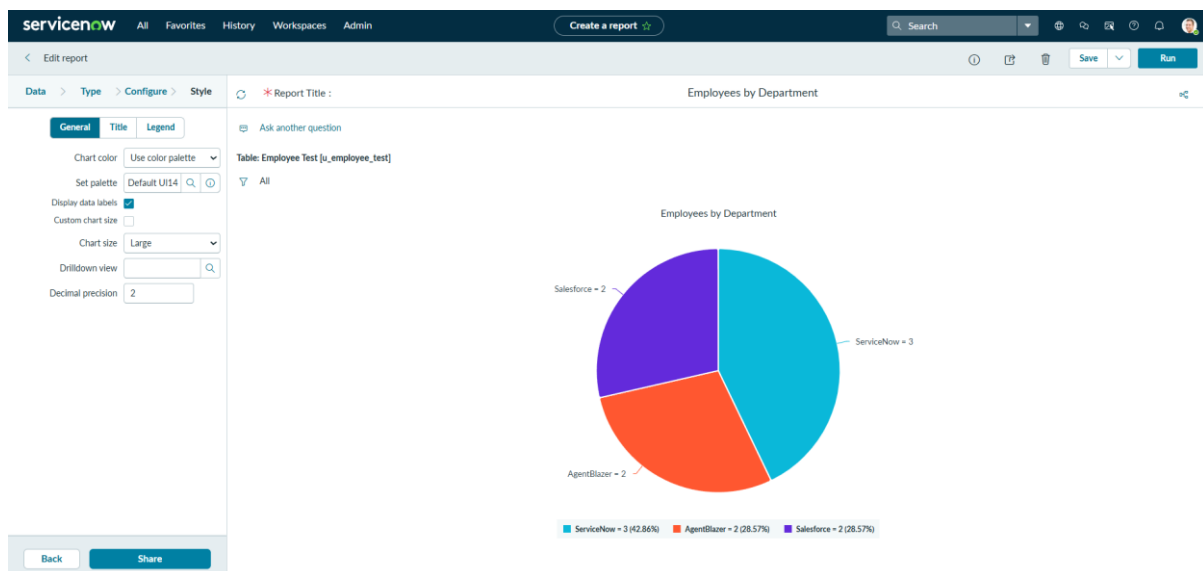


Figure 19: Employees by Department - pie chart

After clicking Run, the pie chart (Figure 19) shows the 7 employees by department: ServiceNow = 3 (42.86%), AgentBlazer = 2 (28.57%) and Salesforce = 2 (28.57%).

Activity 6.2: Report 2 - Employees by Location (Bar Chart)

- Click **New**, enter **Report name:** Employees by Location, **Source type:** Table and **Table:** Employee Test, then click **Next**.
- On the **Type** tab select **Bar** and click **Next**.
- On the **Configure** tab select **Group by:** Location and **Aggregation:** Count, then click **Next**.
- Click **Run** to check the chart and **Save** to store the report.

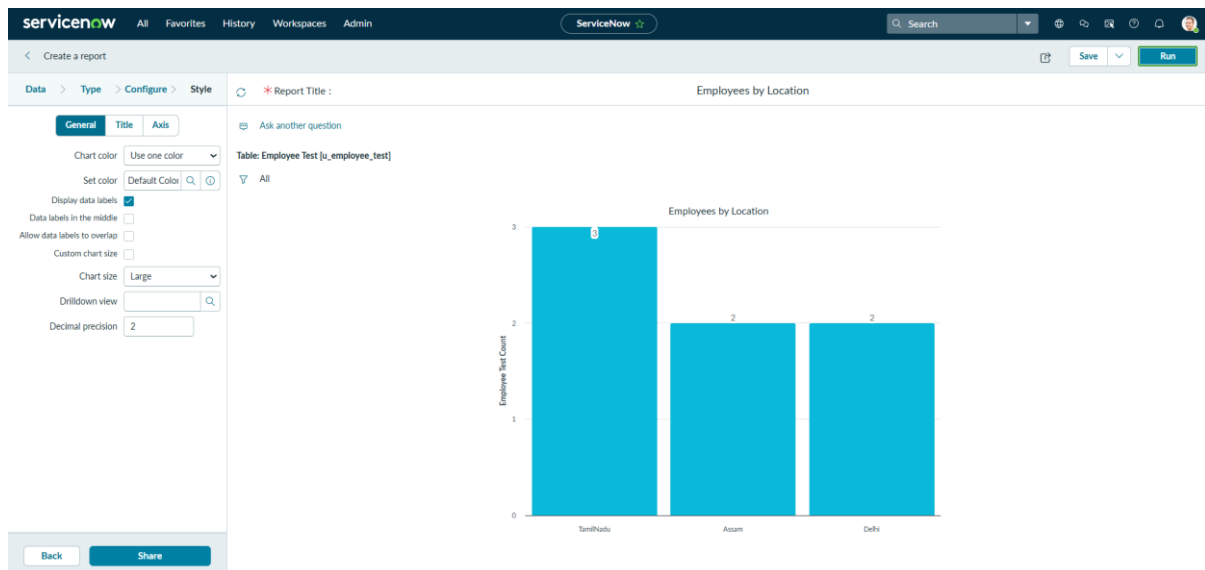


Figure 20: Employees by Location - bar chart

The bar chart (Figure 20) shows TamilNadu = 3, Assam = 2 and Delhi = 2. On the Style tab, Chart color is set to Use one color, Display data labels is ticked and Chart size is Large.

Activity 6.3: Report 3 - Employee List Report

1. Click **New**, enter **Report name:** Employee List Report on the Employee Test table and select **List** as the type.
2. In **Configure**, click **Choose columns** and select Employee ID, Employee Name, Department, Email and Location.
3. Click **Next**, then **Run** and **Save**.

Employee ID	Employee Name	Department	Email	Location
S104083	developer13	Salesforce	developer13@gmail.com	Delhi
S104088	user1	AgentBlazer	user1@gmail.com	Assam
S104082	developer12	Salesforce	developer12@gmail.com	Delhi
S104086	developer5	ServiceNow	developer5@gmail.com	TamilNadu
S104085	developer4	ServiceNow	developer4@gmail.com	TamilNadu
S104089	user2	AgentBlazer	user2@gmail.com	Assam
S104084	developer3	ServiceNow	developer3@gmail.com	TamilNadu

Figure 21: Employee List Report

The list report (Figure 21) shows all 7 records (1 to 7 of 7) with the columns Employee ID, Employee Name, Department, Email and Location.

Milestone 7: Create the Dashboard

Activity 7.1: Create the Dashboard with the In-line Editor

1. Open **All > Dashboards** (or **Platform Analytics > Library > Dashboards**) and click **Create new dashboard**.
2. Select the **In-line editor** tile. The Technical editor opens UI Builder and needs a developer role, so it is not used.
3. Enter **Name:** Employee Analytics Dashboard and a short description, then click **Create new dashboard**.

Activity 7.2: Add the Reports to the Dashboard

1. In the dashboard, click **Add new element** and choose **Data visualization**.
2. In the **Add Visualization** dialog (Figure 22) choose **Saved Visualization** to reuse an existing report, or **New Visualization** to build one from the Employee Test table.
3. Add the pie chart, the bar chart and the Employee List Report one by one.

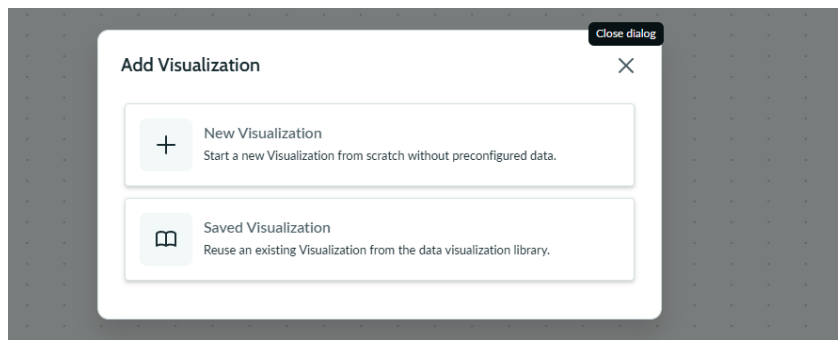


Figure 22: Add Visualization dialog in the dashboard in-line editor

Note: the **Share > Add to Dashboard** option of a report showed "no writable dashboards" for a dashboard created with the in-line editor. The visualizations were therefore added from inside the dashboard using Add new element.

Activity 7.3: Arrange and Save the Dashboard

1. Resize and arrange the three visualizations side by side.
2. Click **Save**. The in-line editor does not save the dashboard automatically.
3. Use the dashboard sharing options to limit access to specific users, groups or roles.

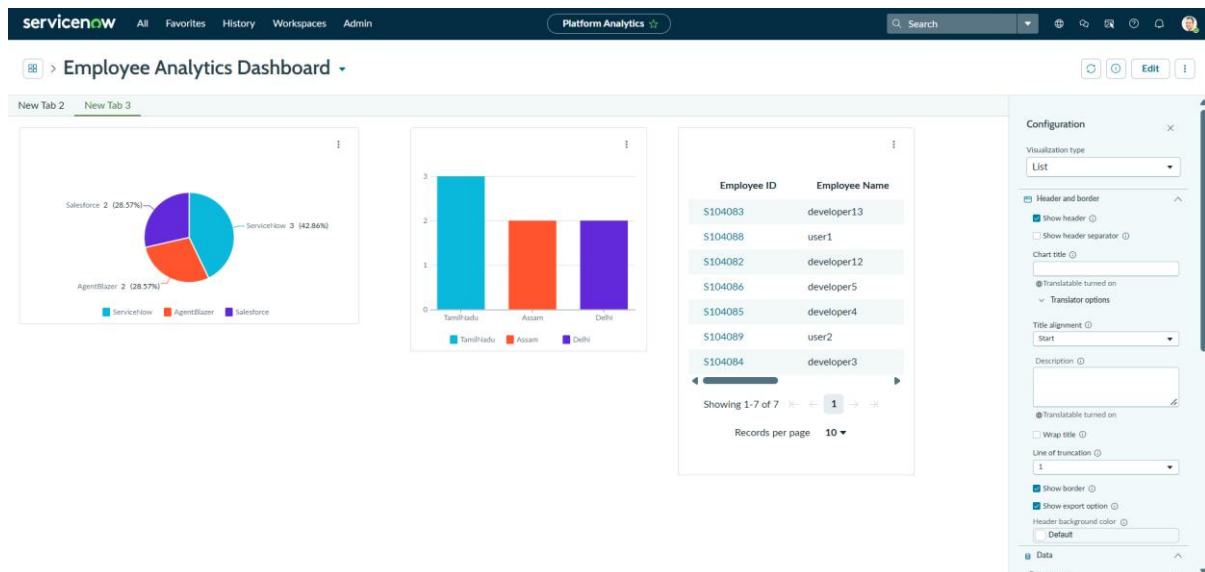


Figure 23: Employee Analytics Dashboard with the three reports

The Employee Analytics Dashboard (Figure 23) shows the pie chart, the bar chart and the Employee List side by side. The screenshot is taken in edit mode with the Configuration panel open for the list element.

10. Running the Application

There are no frontend or backend servers to start. The import is run from the ServiceNow UI:

1. Log in to the instance and open **System Import Sets > Load Data**.
2. Upload the spreadsheet to the staging table and wait for State Complete / Success.
3. Run the transform with the coalesce-enabled map (NMEmp_details Import) and check Transform History.
4. Open **Employee Tests**, the reports or the **Employee Analytics Dashboard** to view the results.

11. API Documentation

The project is operated through the ServiceNow UI. The platform also exposes standard REST APIs for the tables created in this project; these are listed for reference and require an authenticated user with the proper roles.

Purpose	Method	Endpoint	Description
Insert rows into staging table	POST	/api/now/import/u_employee_import	Import Set API: adds a row to the staging table and runs the transform maps.
Read employee records	GET	/api/now/table/u_employee_test	Table API: returns records of Employee Test (supports sysparm_limit, sysparm_query).
Create training record	POST	/api/now/table/u_employee_training_records	Table API: creates a record in Employee Training Records.

Example request:

```
POST /api/now/import/u_employee_import
Content-Type: application/json

{ "u_employee_id": "S104082", "u_name": "developer12", "u_email": "developer12@gmail.com",
  "u_department": "Salesforce", "u_location": "Delhi" }
```

Example response (structure):

```
{ "import_set": "ISET00100xx", "staging_table": "u_employee_import",
  "result": [ { "transform_map": "...", "table": "u_employee_test",
    "status": "updated", "sys_id": "..." } ] }
```

12. Authentication and Access Control

- **Authentication:** users sign in to the ServiceNow instance with a username and password; the browser session is used for all UI actions. REST calls use basic authentication or OAuth.
- **Authorization:** access is controlled by roles and Access Control (ACL) rules. The **admin** role is used to create tables, load data and run transforms.
- **Hr Manager role:** assigned to the user "backup user" so that HR staff can work with employee data (Figure 3 and Figure 4).
- **Dashboards:** any user with an internal role can create an in-line dashboard; the Technical editor needs the ui_builder_admin role. Dashboard sharing can be limited to users, groups and roles.

13. User Interface

S.No	Screen	Description / Figure
1	Table definition	Columns of Employee Training Records (Figure 5)
2	Form Layout	Selected and available fields, split sections (Figure 6)
3	Table form	Employee Training Records record form (Figure 7)
4	Import progress	ImportProcessor result - 5 inserts (Figure 8)
5	Transform Map	Source / target tables and field maps (Figure 9 and Figure 10)
6	Transform progress and history	Transformation complete and Transform History (Figure 11 and Figure 12)
7	Coalesce update run	Run Transform and updated Employee Tests list (Figure 15 and Figure 16)
8	Reports	Pie, bar and list reports (Figure 19 to Figure 21)
9	Dashboard	Employee Analytics Dashboard (Figure 23)

14. Testing

Strategy and tools: the solution was tested manually inside ServiceNow. Each step was verified using the progress page, Import Log, Transform History, the target table lists, the reports and the dashboard.

S.No	Test Case	Expected Result	Observed Result	Status
1	Load spreadsheet into staging table	State Complete, Success	5 processed, 5 inserts, 0 errors (Figure 8)	Pass
2	Run the transform on an import set	Transformation complete	ISSET0010002 completed with Success (Figure 11)	Pass
3	Check Transform History of the first load	Rows inserted, no errors	Total 5, Inserts 5, Errors 0 (Figure 12)	Pass
4	Check field mappings	Source fields mapped to target fields	4 field maps present (Figure 10)	Pass
5	Open Employee Training Records form	Fields shown as per layout	Layout displayed correctly (Figure 7)	Pass
6	Repeated load without coalesce	No duplicate records	Duplicate Employee IDs found (Figure 13)	Issue found (fixed)
7	Update load with Coalesce on Employee ID	Existing IDs updated, new IDs inserted	Updates and inserts applied (Figure 16)	Pass
8	Verify Employee Test after the update	Each Employee ID appears once	7 of 7 records, no duplicates (Figure 16)	Pass
9	Pie, bar and list reports	Reports show the Employee Test data	Reports displayed (Figure 19 to Figure 21)	Pass
10	Dashboard	Three visualizations shown together	Dashboard displayed (Figure 23)	Pass

15. Screenshots and Demo

Figure	Description	Milestone
1	Technical architecture of the data import solution	Section 4
2	ER diagram of the staging, target and user tables	Section 4
3	Role record - Hr Manager	Milestone 1
4	User record with assigned roles	Milestone 1
5	Employee Training Records table with its dictionary entries (columns)	Milestone 2
6	Configuring the Employee Training Records form layout	Milestone 2
7	Employee Training Records form after the layout configuration	Milestone 2
8	Import completed successfully - 5 rows processed, 5 inserted	Milestone 3
9	Transform Map - Employee Transform (source and target tables)	Milestone 4
10	Field Maps (4) of the Employee Transform map	Milestone 4
11	Transformation of import set ISET0010002 completed successfully	Milestone 4
12	Transform History - import set ISET0010009, map NMEmp_details Import (Total 5, Inserts 5, Errors 0)	Milestone 4
13	Employee List Report before coalesce - duplicate Employee IDs (11 records)	Milestone 5
14	Coalesce set to true on the Employee ID field map of NMEmp_details Import	Milestone 5
15	Run Transform - import set ISET0010010 with only NMEmp_details Import selected	Milestone 5
16	Employee Tests after the update - 7 of 7 records, each Employee ID once	Milestone 5
17	Reports module in the navigator (Core UI artifacts)	Milestone 6
18	Report creation - Data tab (Employees by Department)	Milestone 6
19	Employees by Department - pie chart	Milestone 6
20	Employees by Location - bar chart	Milestone 6
21	Employee List Report	Milestone 6
22	Add Visualization dialog in the dashboard in-line editor	Milestone 7
23	Employee Analytics Dashboard with the three reports	Milestone 7

Demo video link:

https://drive.google.com/file/d/1NfjB0lsE0yE642P0S1gS06VRfcII_tXi/view?usp=sharing

16. Known Issues

S.No	Known Issue / Limitation	Workaround / Status
1	Duplicate employee records appeared when the same data was loaded again while Coalesce was false (Figure 13).	Resolved: Coalesce enabled on Employee ID in NMEmp_details Import (Figure 14, Figure 16).
2	Two transform maps (NMEmp_details Import and Employee_Update Import) exist for the same staging table; running both would process every row twice.	Select only the coalesce-enabled map in Run Transform (Figure 15).
3	Share > Add to Dashboard showed "no writable dashboards" for the in-line dashboard.	Resolved: visualizations were added from inside the dashboard using Add new element.
4	Employee Transform (training data) has Coalesce false on all field maps (Figure 10).	Set Coalesce on a unique field if training data is going to be loaded repeatedly.
5	The Employee reference is empty in the sample training record (Figure 7).	Make sure the source Employee value matches a unique field of the User table so the reference can be resolved.
6	Data is loaded manually; no scheduled import is configured.	Planned as a future enhancement.
7	A Personal Developer Instance may hibernate after a period of inactivity.	Wake the instance from the developer portal before use.

17. Future Enhancements

- Configure a **Scheduled Data Import** for recurring automatic loads.
- Add other data sources such as REST or JDBC in addition to Excel and CSV files.
- Add onBefore and field-level transform scripts for data cleansing and validation.
- Enable Coalesce for the training data map and add a dashboard for training records.
- Send email notifications when an import has errors.
- Build a dashboard for import metrics (inserted, updated, ignored and error counts).

18. Conclusion

This project implemented a structured data import solution in ServiceNow using Import Sets and Transform Maps. Spreadsheet data was loaded into a staging table, mapped to custom target tables through field maps and transformed with a verifiable success result in Transform History, giving accurate and efficient migration of employee and training data.

A key enhancement was **Coalesce on Employee ID**. Earlier loads of the same data created duplicate employee records. After Coalesce was enabled, the update file changed the existing employees, added the new employees and left every Employee ID in the Employee Test table exactly once. This keeps the employee data clean, reliable and consistent when data is received from external sources more than once.

The project was extended with a pie chart by department, a bar chart by location and an Employee List Report, which were brought together on the Employee Analytics Dashboard created with the in-line editor. Together with the Hr Manager role for access control, the dashboard gives HR teams a clear, up-to-date view of employee data.

Looking ahead, scheduled imports, additional data sources, transform scripts for cleansing and import-health dashboards can make the solution suitable for larger real-world data migrations.